Fiscal Year:	FY 2016	**Task Last Updated:**	FY 07/14/2016
PI Name:	Gilroy, Simon Ph.D.		
Project Title:	BRIC: Plant Mechanical Signaling During Spaceflight		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Bioregenerative Life Support		
PI Email:	sgilroy@wisc.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	608-262-4009
Organization Name:	University of Wisconsin-Madison		
PI Address 1:	Department of Botany		
PI Address 2:	430 Lincoln Dr.		
PI Web Page:			
City:	Madison	**State:**	WI
Zip Code:	53706-1313	**Congressional District:**	2
Comments:	NOTE: PI formerly at Pennsylvania State University; moved to University of Wisconsin-Madison in 2007 (Info received 7/2009)		
Project Type:	Flight	**Solicitation / Funding Source:**	2012 Space Biology NNH12ZTT001N
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No. of PhD Candidates:		**No. of Master' Degrees:**	
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No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA KSC
Contact Monitor:	Levine, Howard	**Contact Phone:**	321-861-3502
Contact Email:	howard.g.levine@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date should be 5/31/2016 per PI and H. Levine/KSC (Ed., 2/7/17) NOTE: End date is now 12/31/2016 per NSSC information (Ed., 12/14/15) NOTE: End date is now 12/31/2015 per NSSC information (Ed., 1/14/15)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Swanson, Sarah (University of Wisconsin-Madison)		
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Task Description:	Plants are exquisitely sensitive to mechanical stimulation which triggers changes ranging from widespread alteration in patterns of gene expression to profound shifts in development. Indeed, so pervasive is the mechanical stimulation from internal and external forces, such as the weight of tissues and organs, to a biology that has evolved under a constant 1xG, that the resultant signaling is thought to provide a required regulatory background to control a wide range of plant developmental and physiological processes. Despite this acknowledged role in plant morphogenesis, the overall contribution of such mechanical cues to plant function remains poorly defined, largely due to the difficulties in removing the pervasive 1xG-driven mechanical signal when conducting experiments on Earth. The microgravity environment of the International Space Station (ISS) therefore provides a unique tool to both directly address this hypothesis of mechanical load-driven morphogenesis, as well as to ask what features of plant growth in space reflect disruption of this mechano-signaling regulatory network. We have therefore grown seedlings of Arabidopsis thaliana on board the ISS using the Biological Research in Canisters-Petri Dish Fixation Units (BRIC-PDFU) hardware. We are comparing their patterns of growth and gene expression to ground-based controls to ask how the well-defined "fingerprints" of mechanical response are altered in space. In addition, we are characterizing plants grown in space whose mechanical signaling has been artificially activated or inhibited by mutations in a gene (TCH2) which has an established role in the mechanical signaling system of plants. Results from such analyses will address the fundamental question of the role of mechanical loading on plant growth and development on Earth as well as probing how manipulating this system through targeting specific mechanoresponse genes might be used to tailor plant growth to thrive in the unique environment of spaceflight. This experiment has also been supplemented with a parallel GeneLab experiment to monitor the growth, development, and gene expression patterns of a series of natural variants of the plant Arabidopsis thaliana in space. These variants (ecotypes) have been chosen to span the range of likely ecotypes to be used in future spaceflight experimentation and so will provide key baseline comparisons to help understand the impact of ecotypic makeup on responses to spaceflight.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research is addressing how mechanical sensing operates in plants using Arabidopsis thaliana (Mouse Ear Cress). The work targets the Ca2+-dependent signaling machinery and a specific element of this machinery, a gene named touch-2, also known as calmodulin-like 24 (CML24). The patterns of growth and gene expression of wild type and mutants in CML24 grown on the International Space Station (ISS) are being compared to paired plants grown on Earth. On Earth, mechanical signaling systems are known to affect plant vigor, alter their growth habit (favoring plants that are smaller and more robust), and also impact on plant defense responses to pathogens. Therefore, this work aimed at analysis of how Ca2+ signaling is involved in the altered mechanical responses induced during spaceflight should also reveal the role for this system in the more general mechanically-linked responses of plants on Earth. Importantly, this research will allow analysis of the effects of the plant's own weight on its growth and development, research only possible in the microgravity environment of spaceflight. These results should provide molecular targets for potential manipulation to tailor plant growth to varied mechanical environments, especially the relatively mechanically silent environments of spaceflight and controlled enclosed environments on Earth, such as greenhouses. This experiment has also been supplemented with a parallel geneLAB experiment to monitor the growth, development, and gene expression patterns of a series of natural variants of the plant Arabidopsis thaliana in space. The responses of these variants (ecotypes) to spaceflight will not only help further define how the altered environment of space can differentially affect even closely related varieties of plants, but also help provide important background data for the design of future spaceflight experimentation using different ecotypes of this widely adopted model plant.
	This research has used the BRIC-PDFU hardware to address how the altered mechanical environment of spaceflight affects plant growth and gene expression. The research uses mutants in the core touch-response gene CALMODULIN-LIKE 24 (CML24) in the model plant Arabidopsis thaliana and focuses on the Ca2+-dependent signaling network that operates in response to touch stimulation of plants. Both wild type and mutants in CML24 were grown on the International Space Station (ISS), chemically fixed, and then frozen on orbit and then their patterns of gene expression and growth habit compared to parallel ground controls once the samples were returned to Earth. The overall experimental design was to grow seedlings of wild type, cml24-2, and cml24-4 on orbit using the BRIC-PDFU hardware, preserve the seedlings after 8 days using chemical fixation in RNAlater, and freezing in the MELFI. cml24-2 and cml24-4 are two different mutants in the CML24 gene that are thought to continuously activate or inactivate the gene respectively. Upon sample return, whole genome gene expression patterns were assessed using RNAseq to define whether the spaceflight samples showed a transcriptional "fingerprint" of altered mechanical responsiveness, or of other changes in signaling and response networks relative to parallel samples grown under identical conditions on the ground. This experiment was also supplemented with a parallel GeneLab study to monitor the growth, development, and gene expression patterns of a series of natural variants of the plant Arabidopsis thaliana in space. Arabidopsis grows across the planet and subpopulations (ecotypes) show slightly different characteristics dependent on where they originate from. The ecotypes used in the experiment were chosen to span the range of likely ecotypes to be used in future spaceflight experimentation. This approach provides key baseline comparisons to help understand the impact of ecotypic makeup on responses to spaceflight and so provide important background data for the design of future spaceflight experimentation using this widely adopted model plant. Preflight testing (Science and Payload Verification Tests) was successfully completed in 2014 and the payload of 4 BRIC canisters containing a total of twenty 60 mm Petri dishes, each containing 64 seeds, was launched on board SpaceX-CRS4. All on orbit operations were successfully completed and the frozen samples were returned via SpaceX's Dragon capsule, transported to Kennedy Space Center, de-integrated and shipped to the University of Wisconsin-Madison where RNA was successfully extracted and RNAseq transcriptome analysis performed. Six to eight replicate samples were isolated per type of plant. All samples were in the “excellent” range for all mission success criteria. RNAseq was used to monitor the levels of all genes in each sample. To robustly define the up- and down-regulated genes, we used the Tuxedo RNAseq computer analysis pipeline and then applied three separate normalization methods that apply different approaches to statistically assign significantly differentially expressed genes: DESeq, EdgeR, and CuffDiff2. Agreement between multiple analysis approaches provides increased confidence that they are faithfully extracting genes showing true changes in activity in the spaceflight samples. Additionally, we used strict 95% statistical

Task Progress:	significance cutoffs and false discovery rates to maintain stringent selection criteria, and further maximized confidence by looking for where all 3 of the three normalization approaches agreed that a gene was differentially expressed. By these criteria 1556 genes were significantly altered in the cml24 mutant backgrounds with an enrichment in heat shock proteins, that may represent an enhancement of a core spaceflight response system. In addition to alterations in mechanically-responsive genes, flight transcriptomes were seen to parallel those of plants on Earth that have been challenged with the pathogen defense elicitor molecules. Thus, there is significant overlap between the spaceflight and defense elicitor gene sets with, e.g., 19% of the genes altered in flight being associated with bacterial defense response on Earth. As the flight samples were maintained under strict sterile conditions and showed no bacterial or fungal contamination, these results suggest alteration in defense signaling may be one element of the spaceflight response in plants. A further finding from this analysis is that up- and down-regulated flight genes seen in wild-type plants are often even more significantly altered in the cml24-4 mutant (the loss of function mutant in the CML24 gene) and this response can be reversed in the cml24-2 mutant that shows irreversible activation on the CML24 gene. Overall what emerges from these analyses is a significant enrichment in defense- and mechanically-responsive genes in the spaceflight environment. Parallel analysis of the multiple ecotypes flown under the Genelab component of this experiment indicated genes encoding heat shock proteins are up-regulated in response to spaceflight. Heat shock proteins are molecular chaperones that act in many stress response pathways in addition to temperature stress. At a cellular level, these proteins act to refold or remove other proteins damaged by environmental conditions and have been highlighted in previous spaceflight responses of Arabidopsis cell cultures. Further analysis indicates other related molecular chaperones that change in some, but not all, of the ecotypes, further emphasizing the link between chaperones and response to spaceflight. Importantly, data logger information from BRIC-19 showed that no thermal excursions from the ISS ambient of ~24°C occurred during the experiment indicating these transcripts are not simply reporting a heat shock event but suggesting upregulation of a broad stress-related machinery in response to some other aspect of the spaceflight environment. Spaceflight also led to down-regulation of classes of genes and prominent amongst these were enzymes of the peroxidase family that are involved in remodeling the plant cell wall. A role for these enzymes in spaceflight was previously highlighted in results from the BRIC16 experiment, suggesting alteration in cell wall structure may be a commonly occurring plant response to the spaceflight environment. Gene ontology/classification analyses of these data further suggest response to oxidative stress is a major component of the core spaceflight stress network, a conclusion supported by gene network analysis, such as with the QTLNetminer tool. The raw gene expression data for these analyses has been deposited as GeneLab Data System-37. In collaboration with Dr. Richard Morris at the John Innes Centre, Norwich, UK, work has also been conducted on modeling how reactive oxygen species (ROS) and Ca2+ interact to propagate stress responses in plants. Using a combination of in vivo imaging of Ca2+ and ROS dynamics and analysis of responses in the AtRBOHD mutant (defective in the Respiratory Burst Homolog D enzyme responsible for producing many signaling-related changes in ROS), the role of this enzyme in ROS-assisted Ca2+-dependent stress signaling has been dissected. This quantitative analysis allowed Richard Morris to develop a math-based model to describe this phenomenon. The model predicts a role for AtRBOHD-dependent extracellular ROS production in cell-to-cell movement of stress information within the plant. Using measurements of responses to inhibitors of AtRBOHD, mutants, and quantitative imaging, the predictions of the model have been tested and the model validated.

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